G-Synth: Small Sequence Design and Extended Sequence Design for auditable synthesis-ready DNA and post-sequencing validation

Software development and protein-to-sequence validation with PCR-free insulin glargine A/B-chain constructs

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Abstract

Nucleic-acid design for synthesis is often fragmented across codon tools, restriction maps, plasmid editors and sequencing viewers, complicating proof that an ordered molecule is the one later cloned and assessed. We developed G-Synth, an open-source application and dependency-free Python engine implementing 15 source-traceable host codon profiles and two synthesis-centred workflows: Small Sequence Design (SSD) for one complementary oligonucleotide pair and Extended Sequence Design (ESD) for tiled, directional pairs. Both derive strand-specific restriction products and block release unless in-silico annealing and ordered ligation reconstruct both intended strands exactly. The implementation separates the tested molecular engine, a Django REST service and a React/TypeScript interface. We first exercised the current workflow from the mature insulin glargine A- and B-chain amino-acid sequences. G-Synth preserved both non-methionine N termini, reverse-translated them with the selected Escherichia coli profile, applied synthesis and NdeI/XhoI constraints, generated SSD order molecules, verified their duplexes and produced clonable 5,490-bp and 5,523-bp pET-21a(+) models. These new synonymous DNA designs were validated computationally. In a separate retrospective experimental track, SSD regenerated the constructs synthesized at the bench; all four oligonucleotides matched their order records exactly. G-Synth then assembled the four author-designated chromatograms into 100%-covered, 100%-identical consensus sequences. Bidirectional overlap was 56.1% for A and 71.2% for B, with 100% F/R agreement; independent Biopython analysis reproduced complete coverage and identity. The engine, API and interface passed 1,113, 265 and 92 tests, respectively. The exact SSD/ESD release-gate combination was not identified in prior design-automation literature and is, to our knowledge, first automated and disclosed here. Protein expression, folding and bioactivity were not tested.

Keywords: synthetic biology; gene synthesis; restriction cloning; Sanger sequencing; software validation; insulin glargine.

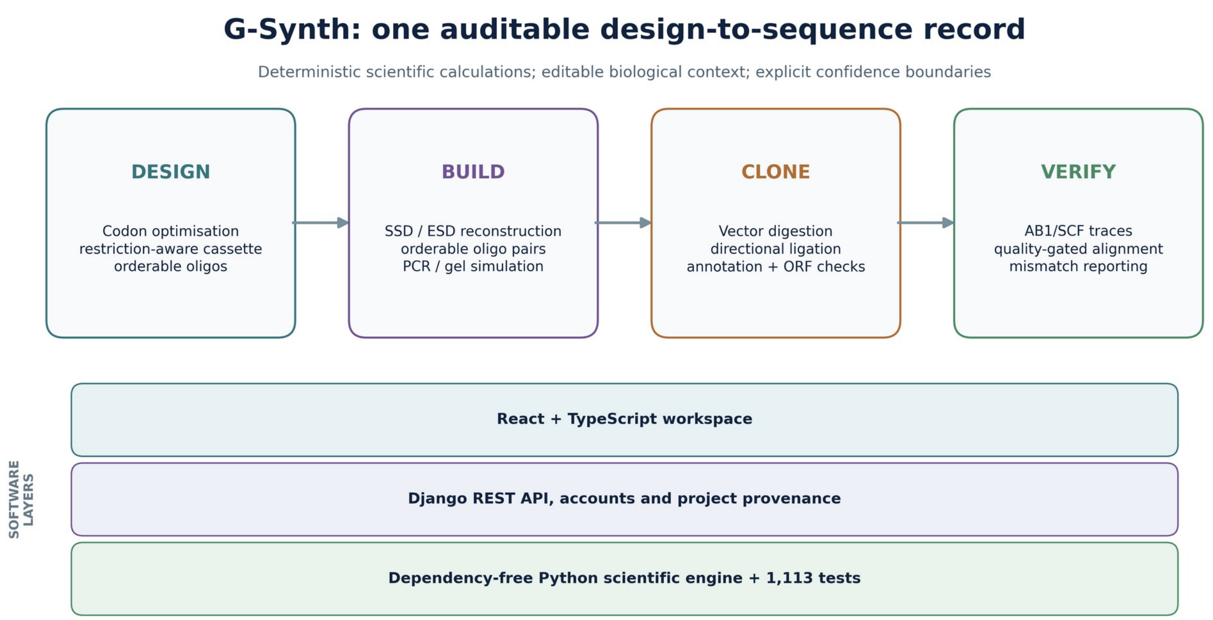

Figure 1. G-Synth workflow and software architecture. The design, build, clone and verify stages share a deterministic Python engine. The API stores project provenance and the web workspace renders the engine outputs without reimplementing biological calculations.

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1. Introduction

Synthetic-biology design–build–test–learn cycles depend on an unbroken relationship between a digital construct and the molecules handled at the bench [13,58–60]. GeneDesign automates synthetic-gene design [61,62], j5 optimizes DNA-assembly plans [58], basicsynbio couples BASIC constructs to transferable build instructions [59], and DNA-BOT links a standardized scheme to laboratory automation [60]. Geneious provides broad commercial cloning and sequence analysis [65], whereas DIVA, CloneCoordinate and InSillyClo extend design–implementation–verification or campaign-level construction planning [69–71]. These systems establish that G-Synth cannot credibly be called the first general end-to-end DNA-construction platform. A distinct gap nevertheless remains for laboratories that order short coding constructs as complementary oligonucleotides: restriction sites may be treated as literal tails instead of strand-specific cut geometry; an enzyme-retained start codon may be duplicated; a reverse molecule may receive the wrong cohesive-end remainder; or an apparently plausible chromatogram may be reported as complete verification despite incomplete evidential support.

These problems require a workflow in which each exported sequence is derived from explicit molecular rules and then re-read from the simulated duplex. G-Synth therefore defines SSD for constructs released as one forward/reverse synthesis pair and ESD for longer constructs decomposed into multiple hybridized pairs joined by unique, directional internal overhangs. Their shared end-to-end unit is a traceable molecular record: biological target → synthesis-ready order molecules → reconstructed duplex → recombinant plasmid → annotated construct → sequencing evidence. The engine derives enzyme products from recognition and cut positions, constructs the full coding cassette, creates every strand, re-anneals and re-ligates the strands in silico, checks the outer ends and coding frame, simulates vector digestion and ligation, and maps capillary reads back to the same reference.

Insulin glargine A- and B-chain constructs offered both a current end-to-end demonstration and a stringent retrospective case study. The analysis therefore used two explicitly separated tracks. First, G-Synth began from the mature amino-acid sequences, performed E. coli-conditioned reverse translation and carried the resulting synonymous DNA through SSD, hybridization and restriction cloning. Second, the nucleotide inputs and four molecules synthesized, hybridized, cloned into pET-21a(+), transformed into Escherichia coli DH5α and sequenced in the source study were reanalysed without substitution. Benchling had supplied reverse-complement operations and Geneious Prime the visual cloning and alignment checks in that study; G-Synth is the primary system in the present analysis. The objectives were to (i) demonstrate the complete current protein-to-clone workflow; (ii) reproduce the exact experimental order molecules; (iii) validate terminal-end geometry, junctions, open reading frames and plasmid context; (iv) assemble the physical F/R trace evidence while distinguishing consensus coverage from bidirectional overlap; and (v) compare G-Synth's focused, auditable workflow with established software categories.

2. Software development, implementation and scientific logic

2.1 Architecture and development tools

G-Synth was implemented as three deliberately separated layers (Figure 1). The gsynth_engine package is a dependency-free Python 3.11+ scientific core containing sequence normalization, codon optimization, nearest-neighbour thermodynamics, enzyme definitions, duplex construction, multi-oligonucleotide assembly, PCR, cloning, ligation, annotation, gel simulation and read verification. A Django 5.2/Django REST Framework service validates JSON requests, applies JWT-based account isolation and serializes immutable calculation outputs with project provenance; SQLite supports local development and PostgreSQL is required for production deployment. A React 18 and TypeScript 5 single-page workspace, compiled with Vite, renders engine outputs and uses SeqViz for sequence maps without reimplementing molecular calculations. Production serving uses Gunicorn and WhiteNoise. Pytest, property and golden-reference tests validate the engine and API; Vitest, Testing Library, TypeScript compilation and production builds validate the interface; Ruff enforces Python static quality. Biopython is used only for independent file-format and alignment cross-checks, not as a hidden runtime dependency of the engine [66]. This architecture follows robustness, workflow-readiness and research-software traceability recommendations [72–74].

Codon optimization is explicitly conditioned on the selected expression host. Fifteen profiles cover bacterial (Escherichia coli, Bacillus subtilis, Pseudomonas putida, Lactococcus lactis, Corynebacterium glutamicum and Streptomyces coelicolor), yeast (Saccharomyces cerevisiae, Komagataella phaffii, Kluyveromyces lactis and Yarrowia lipolytica), mammalian (human and Chinese-hamster species proxies), insect (Spodoptera frugiperda and Drosophila melanogaster species proxies) and plant (Nicotiana benthamiana) systems. Each profile preserves all 64 raw codon counts, NCBI taxon, source dataset, sample size, GC and snapshot checksum. Counts from the September 2021 FDA HIVE-CUTs/CoCoPUTs release are normalized within each synonymous family [83,84]; RefSeq species aggregates are preferred, with a disclosed GenBank fallback when that snapshot lacks a RefSeq table. Every candidate is translated after optimization, and amino-acid identity is a blocking invariant. A species-wide result is labelled profile-relative CAI rather than strict CAI or predicted yield; a documented highly expressed reference-gene set can override it for strain-, tissue- or cell-line-specific analysis [82,85–87].

The same workflow accepts a peptide and performs host-conditioned reverse translation. Its start logic distinguishes the coding ORF from a mature or post-translationally processed product [88–90]. Automatic mode treats an N-terminal methionine as an existing initiator and otherwise preserves the supplied peptide as a mature product whose initiation is provided upstream by Design. Because sequence alone cannot establish biological maturity, the user may override this inference. Declaring a non-M peptide as a complete expression protein adds exactly one initiator methionine/ATG; declaring it mature never changes its residue sequence. The engine records the input peptide, resolved role and any added initiator, re-translates the generated DNA and transfers the corresponding coding-state flag to Design. It does not predict in-vivo methionine excision.

2.2 Restriction enzymes as cut geometry

An enzyme is represented by its recognition sequence and top- and bottom-strand cut offsets, not by a prewritten tail. From these offsets the engine derives overhang sequence, length and 5′/3′ polarity for either end of a linear fragment. This matters because the bases that remain on a forward oligonucleotide depend on whether an enzyme is placed at the left or right end and on which strand is ordered. The selectable table contains 109 non-redundant in-site cut geometries representing 289 commercially available enzyme names from REBASE; isoschizomers remain searchable aliases rather than duplicate map sites [75]. Every geometry is exercised in both terminal positions. HindIII, for example, is rendered from A|AGCTT / TTCGA|A cut coordinates and therefore appears with the correct 5′ cohesive end rather than being lost by a recognition-site-only search. Enzymes with ambiguous sites, out-of-site cleavage or two cuts are outside this model and are not claimed as supported. The enzyme-table checksum is stored in provenance records.

2.3 Coding cassette and PCR-free oligonucleotide design

In Small Sequence Design (SSD), the default expression cassette is ATG (unless the retained enzyme product supplies it), a flexible linker, 6×His tag, a second linker, a thrombin-recognition sequence and the user insert. A stop codon can be retained or added according to the intended downstream fusion. G-Synth checks translation, internal selected-enzyme sites, sequence composition, repeats and secondary-structure warnings. Melting temperatures use nearest-neighbour thermodynamics under recorded ionic and strand-concentration assumptions [80], not a GC-only formula.

Within the SSD single-pair length limit, G-Synth generates one forward and one reverse strand with the exact complementary core and asymmetric terminal cohesive ends. Extended Sequence Design (ESD) divides longer constructs into forward/reverse hybridized fragments joined through orthogonal 4–8-nt internal junctions. Every SSD or ESD proposal is simulated end to end: each pair is annealed; ESD fragments are ligated in the declared order; both reconstructed strands must equal the target; every internal overhang must be unique against its reverse complement; and the two outer products must equal the selected restriction-enzyme ends. Export remains blocked if any invariant fails.

2.4 Hybridization as the gate between design and cloning

Alignment and hybridization answer different questions and are therefore presented as separate modes in one workspace. Alignment may introduce gaps to compare sequence similarity; hybridization instead tests an ungapped antiparallel placement of two molecules entered 5′→3′. G-Synth reverse-complements the second input only for the physical drawing, marks every complementary pair and mismatch, and leaves terminal displacement visible as a strand-specific 5′ or 3′ overhang. The simple view summarizes the duplex core and both ends, whereas the detailed view shows every base, coordinate and polarity. Melting temperature is reported only for a perfectly matched overlap under explicit strand, Na^{+}, Mg^{2+} and temperature conditions; it is withheld for mismatched duplexes because the implemented nearest-neighbour model does not include mismatch parameters [80].

A verified SSD or ESD design can be transferred directly to this view with its left and right enzyme identities. Editing either enzyme invalidates the inherited molecules and requires redesign, preventing a sequence from being relabelled with ends it does not possess. Only an exactly complementary duplex can continue to cloning. This implements a visible state transition from an orderable pair, through simulated annealing, to an insert whose exposed ends can be compared with the independently digested vector.

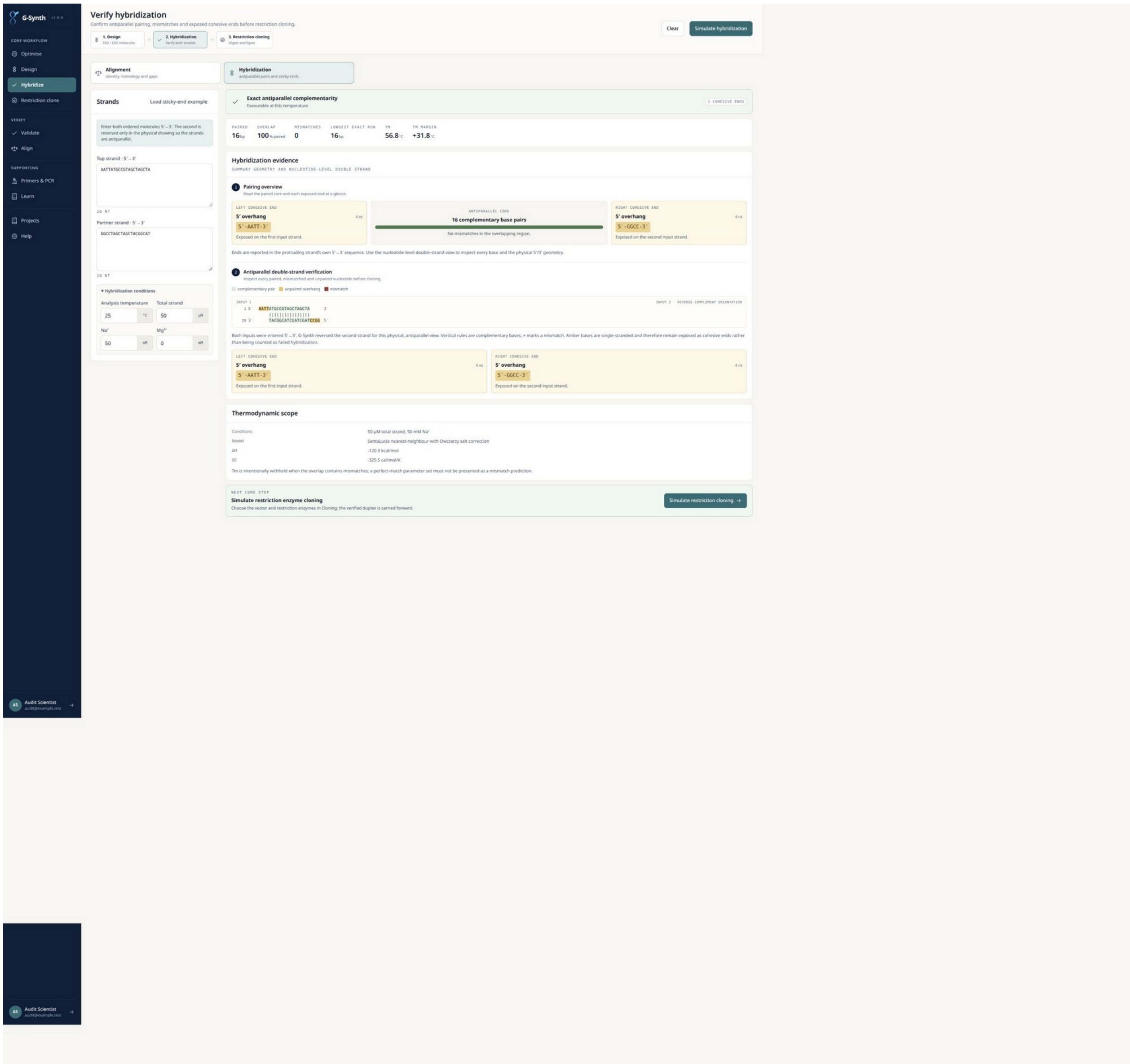

Figure 2. G-Synth hybridization view after direct transfer from SSD. Both order molecules remain entered 5′→3′; the lower strand is drawn antiparallel. The 123-bp core is completely paired, while the NdeI-derived 5′-TA and XhoI-derived 5′-TCGA products remain visibly unpaired as cohesive ends.

Alt text: G-Synth hybridization view after direct transfer from SSD. Both order molecules remain entered 5′→3′; the lower strand is drawn antiparallel. The 123-bp core is completely paired, while the NdeI-derived 5′-TA and XhoI-derived 5′-TCGA products remain visibly unpaired as cohesive ends.

2.5 Cloning, annotation and virtual experiments

The cloning module imports FASTA, GenBank and SnapGene DNA files and includes verified pET-21a(+) and pET-21(+) sequences. It checks vector identity, requires each selected enzyme to cut once and presents the workflow as Vector + duplex Insert → Product. Before ligation, six compatibility checks report strand pairing, unique vector cuts, orientation, site regeneration and reading-frame consequences. Simple and detailed junction views place each exposed insert end beside the complementary vector end; the joined sequence is explicitly labelled as an expected product until the action is performed. The product map, diagnostic gel and exports remain hidden until the user selects Ligate compatible ends. Only then does an accessible green confirmation panel display the verified recombinant length, the Vector + Insert → Product relationship and close-up nucleotide sequences for both closed junctions. This state transition simulates joining the phosphodiester backbone and is not represented as an experimental ligation. The coordinate-level annotated view handles circular-origin crossings, strand direction and codon-aligned translation. Curated exact matches to common promoters, operators, tags, linkers and cleavage motifs are suggestions, not silently accepted claims; users can create, name, move or delete features and preserve edits in GenBank export.

PCR simulation separates a primer's annealing 3′ segment from its intentionally unpaired 5′ extension. The extension becomes part of the amplicon only after polymerase copying, a distinction shown explicitly in the hybridization view. Diagnostic digests produce fragment sizes from the simulated molecule, and virtual gels position those products relative to explicit 100-bp, 1-kb and broad-range marker lists. These gels are labelled as in-silico predictions because migration, topology, partial digestion, staining and band intensity are experimental properties.

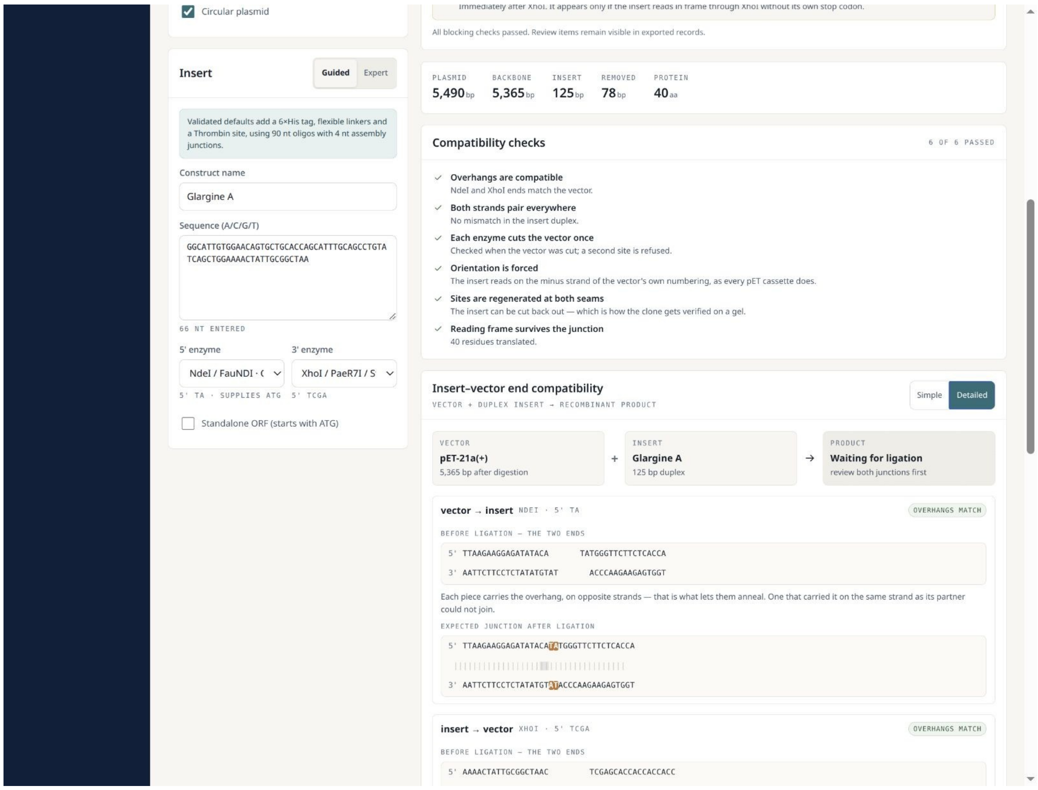

Figure 3A. Pre-ligation compatibility gate for the current glargine A design. G-Synth shows the 125-bp insert, the NdeI and XhoI vector–insert junctions, their opposite polarities and all six passed checks while labelling the joined sequence as expected and withholding recombinant-product analyses.

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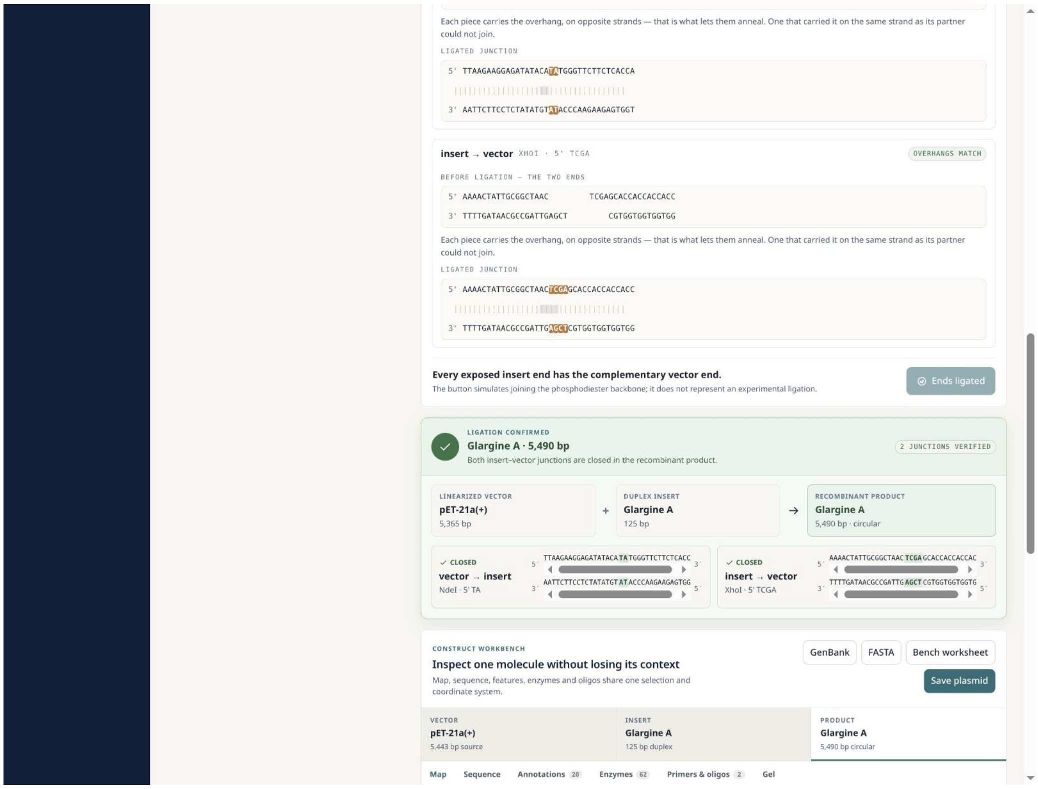

Figure 3B. Post-ligation state for the current glargine A-chain construct. A green confirmation panel reports the 5,490-bp recombinant product and shows both closed nucleotide junctions only after explicit in-silico ligation; the plasmid map, diagnostic digest, gel and exports then become available.

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2.6 Quality-aware capillary-trace verification

G-Synth detects ABI/ABIF and Standard Chromatogram Format (SCF) content, reads called bases, qualities, peak positions and the four fluorescence channels, evaluates both orientations and aligns reads to a linear or circular reference. Forward and reverse calls are transformed into reference orientation and assembled position by position; equal-support conflicts remain N rather than being resolved in favor of the reference. The interface reports assembled-consensus coverage, consensus identity, the fraction supported by both orientations and F/R agreement within that overlap as distinct quantities. Reference-aligned chromatograms display consensus, strand, base qualities, four-channel peaks and mismatches. Optional user-selected quality trimming is retained for prospective workflows, but quality thresholds must be predefined for the relevant instrument, basecaller and study rather than imposed retrospectively. Phred-style quality values are treated as estimates of base-call error probability [78,79].

2.7 Originality and application scope

The originality of G-Synth is the formalization and automation of SSD and ESD, not the invention of restriction cloning, codon optimization, oligonucleotide synthesis or Sanger sequencing individually. GeneDesign, j5, DNA Chisel, basicsynbio, DIVA, CloneCoordinate and InSillyClo provide important but differently scoped automation [58–63,69–71]. In the literature reviewed through 2 September 2026, no system was found that defines SSD and ESD as synthesis-order workflows and requires exact reconstruction of both intended strands from their exported molecules before release, while carrying the same hashed design through restriction cloning and F/R capillary consensus. We therefore describe G-Synth as the first disclosed automation of these specifically defined workflows, to our knowledge, rather than as the first general DNA-design or end-to-end construction platform.

This focus supports applications in which small or modular DNA constructs must move rapidly from concept to synthesis: therapeutic-peptide precursors; affinity-tag and protease-cleavage cassettes; vaccine or diagnostic antigens; peptide standards and positive controls; recombinant growth factors, cytokine fragments and antimicrobial peptides; enzyme domains; regulatory elements; reporter constructs; and modular synthetic-biology parts. The glargine A/B case study is especially relevant because therapeutic peptides impose simultaneous constraints on exact amino-acid identity, purification architecture, cleavage logic, reading frame and post-cloning verification. G-Synth addresses the DNA design-to-sequence portion end to end. Expression yield, processing, oxidative folding, higher-order structure, potency, safety and regulatory comparability remain separate experimental stages.

3. End-to-end G-Synth workflow from insulin glargine protein sequences

3.1 Protein input, biological role and host-conditioned reverse translation

The current G-Synth workflow began with the amino-acid sequences of the mature insulin glargine chains: GIVEQCCTSICSLYQLENYCG for A and FVNQHLCGSHLVEALYLVCGERGFFYTPKTRR for B. Neither input begins with methionine because each represents a mature chain or precursor-derived chain segment rather than a complete standalone translation product. Automatic peptide-role inference therefore resolved both as mature peptide/fusion inserts, preserved every supplied residue and did not introduce an artificial N-terminal methionine. This decision remained user-overridable and was recorded in the output provenance.

Escherichia coli was selected as the intended expression host. G-Synth reverse-translated each peptide with the committed September 2021 HIVE-CUTs/CoCoPUTs species profile while excluding internal NdeI and XhoI sites, low-frequency profile codons, homopolymers longer than five bases, repeats of 15 nt or longer and 50-nt windows outside 40–60% GC. A TAA stop codon was retained. Translation of each candidate was then compared with the supplied peptide plus termination as a blocking invariant. Both sequences passed without warnings: A was 66 bp, with profile-relative CAI 1.000 and 50.0% GC; B was 99 bp, with profile-relative CAI 0.944 and 53.5% GC. These values describe fit to the selected species-wide codon profile and are not predictions of expression yield.

Chain	Protein input	G-Synth E. coli-conditioned DNA (5′→3′)	Profile fit
A	GIVEQCCTSICSLYQLENYCG	GGCATTGTGGAACAGTGCTGCACCAGCATTTGCAGCCTGTATCAGCTGGAAAACTATTGCGGCTAA	CAI 1.000; GC 50.0%
B	FVNQHLCGSHLVEALYLVCGERGFFYTPKTRR	TTTGTGAACCAGCATCTGTGTGGCAGCCATTTAGTGGAAGCGCTGTATCTGGTGTGCGGCGAACGCGGCTTCTTTTATACCCCGAAAACCCGCCGCTAA	CAI 0.944; GC 53.5%

3.2 SSD, duplex verification and restriction-cloning simulation

Each newly back-translated sequence was transferred directly to SSD as a non-standalone insert. G-Synth placed it after the NdeI-supplied initiator and the MGSSHHHHHHSSGLVPRGS purification/linker/thrombin cassette, then generated the strand-specific NdeI and XhoI products. The A design contained 125-nt forward and 127-nt reverse order molecules; B contained 158-nt forward and 160-nt reverse molecules. In both cases, simulated antiparallel annealing reconstructed the full duplex core exactly and left only the expected 5′-TA and 5′-TCGA cohesive ends exposed. The verified duplexes passed the vector-cut, end-compatibility, orientation, reading-frame and site-regeneration gates and produced 5,490-bp and 5,523-bp pET-21a(+) models after explicit in-silico ligation.

The G-Synth DNA sequences differ synonymously from the physically synthesized experimental constructs while encoding the same peptides. They therefore demonstrate the current protein-to-clone workflow and its computational invariants but were not the molecules tested at the bench. Experimental validation was assigned only to the preserved experimental-sequence track in Section 4. This separation prevents retrospective bench data from being presented as physical validation of newly generated codon choices. Exact new order molecules, hashes and junction records are supplied in the Supporting Information and machine-readable evidence.

3.3 Software and molecular validation framework

Validation combined unit, property, API and interface tests with a retrospective experimental case study. The engine suite contains golden examples, error-path tests, host-profile invariants, peptide-start decisions, enzyme-wide properties and antiparallel-hybridization cases covering 5′/3′ overhangs, blunt ends, ambiguity, mismatches and thermodynamic reporting boundaries. The defining assembly property is equality between the intended construct and both strands reconstructed from the exported oligonucleotides. API tests cover request validation, host selection, peptide-to-DNA serialization, authentication, projects, sequence import, hybridization, PCR and security. Interface tests cover the design and validation views, host-profile and peptide-role selection, automatic Design-to-Hybridization-to-Restriction-cloning handoff, simultaneous compact and nucleotide-level duplex evidence, editable annotations, primer-tail hybridization, gels, Learn content, accessible ligation confirmation and responsive behaviour. The complete suites were executed from the manuscript working tree on 2 September 2026: 1,113/1,113 engine, 265/265 API and 92/92 interface tests passed.

Validation layer	Primary question	Result
Molecular engine	Do algorithms preserve molecular invariants and reject invalid inputs?	1,113 passed
HTTP/API	Are calculations exposed consistently with accounts, projects and security controls?	265 passed
Web interface	Are outputs rendered and interactions retained without reimplementing biology?	92 passed
De-novo glargine workflow	Do protein inputs preserve identity through host optimization, SSD, hybridization and cloning?	A and B passed computational gates
Experimental glargine design	Do the recorded A/B inputs regenerate the four molecules synthesized at the bench?	4/4 exact
Cloning simulation	Are NdeI/XhoI ends, junctions and ORFs coherent in pET-21a(+)?	A and B passed
Sequencing evidence	Do oriented F/R reads assemble across each insert and agree in their overlap?	100% consensus coverage and identity; 100% overlap agreement

4. Retrospective validation with experimentally tested glargine constructs

4.1 Experimental nucleotide inputs and cassette logic

For experimental traceability, G-Synth next received the exact optimized nucleotide inputs preserved in the experimental record rather than substituting the newly generated synonymous sequences from Section 3. The tested A-chain insert was 66 bp and encoded GIVEQCCTSICSLYQLENYCG; the tested B-chain insert was 99 bp and encoded FVNQHLCGSHLVEALYLVCGERGFFYTPKTRR. The second sequence includes the two C-terminal arginine residues characteristic of insulin glargine's B-chain precursor context. For each construct, G-Synth placed the insert after MGSSHHHHHHSSGLVPRGS, retained TAA termination and selected NdeI at the 5′ end and XhoI at the 3′ end. NdeI's retained bases supply the initiating ATG, so adding a separate ATG would duplicate the start. The resulting products were MGSSHHHHHHSSGLVPRGSGIVEQCCTSICSLYQLENYCG and MGSSHHHHHHSSGLVPRGSFVNQHLCGSHLVEALYLVCGERGFFYTPKTRR.

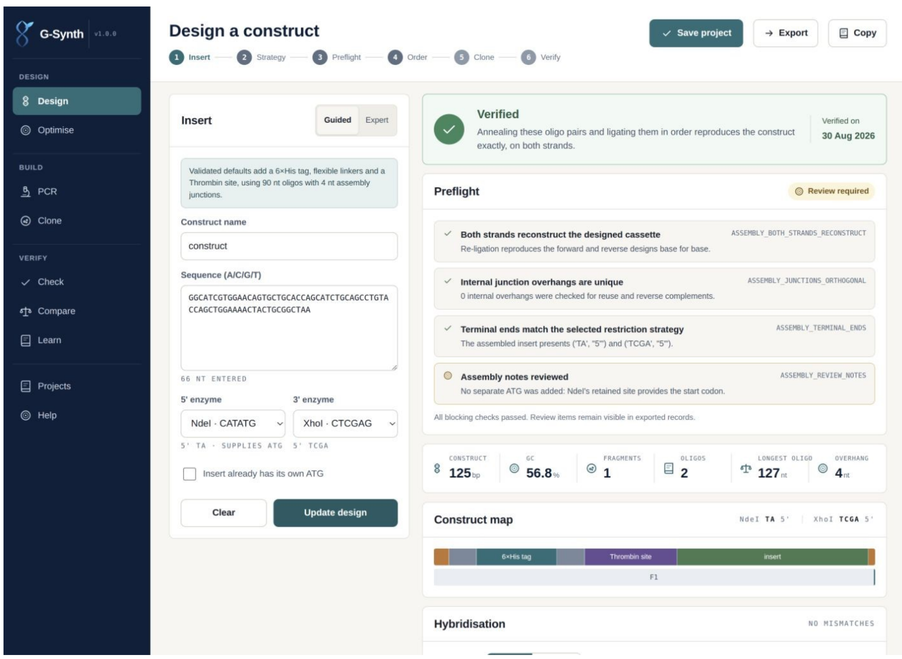

Figure 4. G-Synth design view for the glargine A-chain input. The release gate reports exact two-strand reconstruction, terminal-end compatibility and the NdeI start-codon note before displaying the orderable construct.

Alt text: G-Synth design view for the glargine A-chain input. The release gate reports exact two-strand reconstruction, terminal-end compatibility and the NdeI start-codon note before displaying the orderable construct.

Metric	A-chain construct	B-chain construct
Optimized chain insert	66 bp	99 bp
Forward / reverse oligo	125 / 127 nt	158 / 160 nt
Forward / reverse GC	56.8 / 57.5%	57.0 / 57.5%
Forward / reverse Tm	83.4 / 84.1 °C	84.6 / 85.1 °C
Observed terminal products	5′ TA / 5′ TCGA	5′ TA / 5′ TCGA
Exact match to experimental synthesis record	Forward and reverse	Forward and reverse

Metric	A-chain construct	B-chain construct
Recombinant pET-21a(+) length	5,490 bp	5,523 bp
Junction restriction sites	NdeI and XhoI regenerated	NdeI and XhoI regenerated

The reverse oligonucleotide is two nucleotides longer than the forward molecule in each pair (127 versus 125 nt for A; 160 versus 158 nt for B). This is an expected consequence of the strand-specific cohesive-end remainders, not a discrepancy. All four generated sequences were identical to the corresponding experimental order records, and the complementary cores reconstructed without mismatch.

4.2 pET-21a(+) cloning and feature-level validation

G-Synth digested the bundled pET-21a(+) sequence with NdeI and XhoI, checked the insert's physical ends and simulated directional ligation. Both junctions were compatible and regenerated their recognition sites. The A and B recombinant molecules were 5,490 bp and 5,523 bp. Translation from the NdeI-supplied ATG reached the intended TAA without frameshift. Because the inserts terminate before the vector's downstream coding region, the pET-21a(+) C-terminal His tag is not translated; the designed N-terminal 6×His cassette remains present. Similarly, cloning at NdeI replaces the vector's optional N-terminal T7 tag with the designed cassette. These are sequence consequences, not predictions of expression, solubility, cleavage efficiency or biological activity.

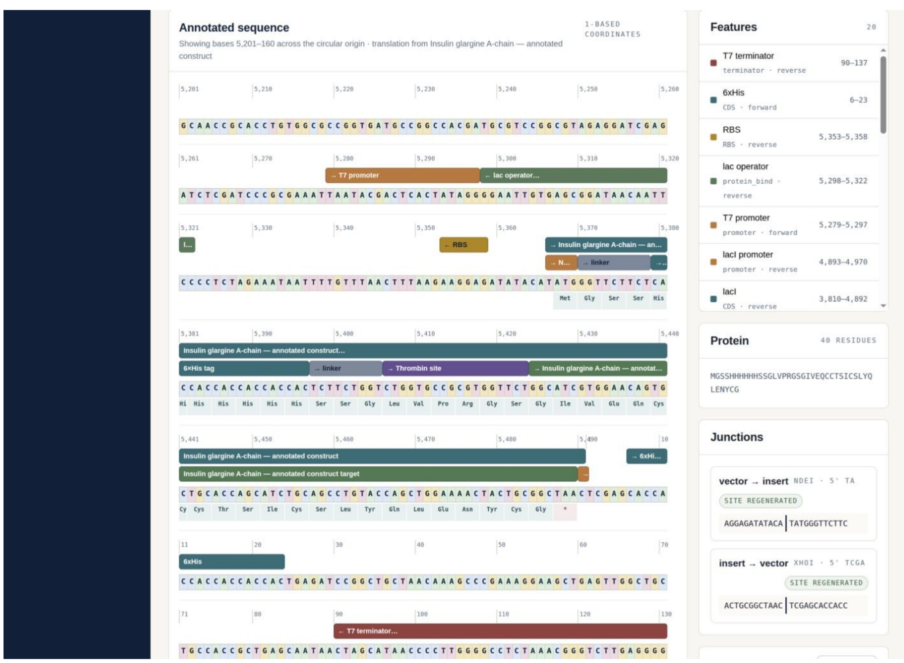

Figure 5A. G-Synth coordinate-level view of the insulin glargine A-chain recombinant pET-21a(+) construct. Editable feature tracks, codon-aligned translation and the regenerated NdeI and XhoI junctions are shown across the circular origin.

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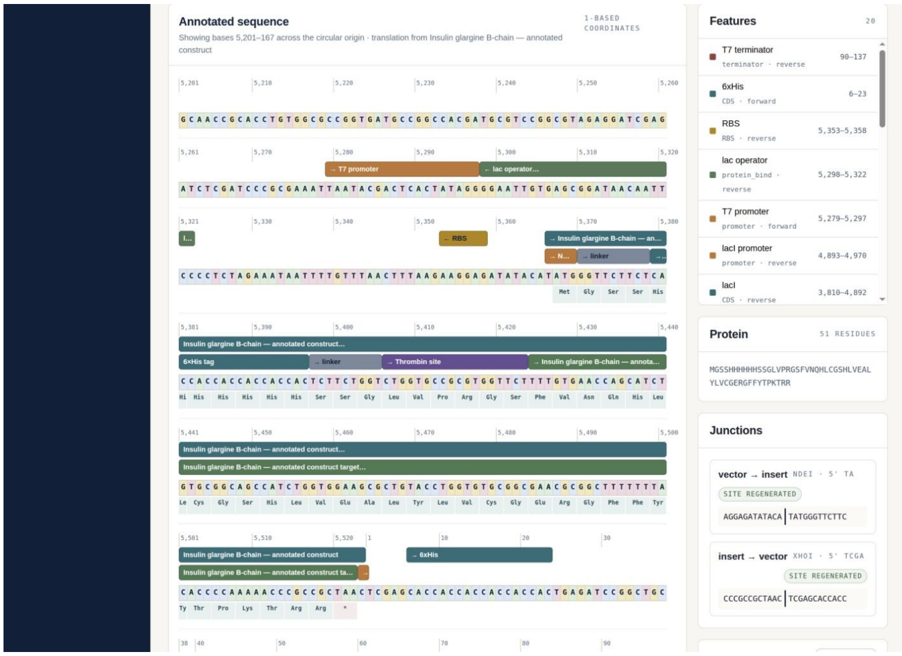


Figure 5B. G-Synth coordinate-level view of the insulin glargine B-chain recombinant pET-21a(+) construct, using the same editable annotation and strand-aware coordinate system.

Alt text: G-Synth coordinate-level view of the insulin glargine B-chain recombinant pET-21a(+) construct, using the same editable annotation and strand-aware coordinate system.

4.3 Experimental synthesis and cloning evidence

The retrospective data originated from an independent wet-laboratory project at the Genomics Technology Platform of the Higher School of Biological Sciences of Oran. Forward and reverse strands were synthesized on a MerMade 4 instrument at 1 µmol scale, purified, and annealed in 2× SSC with 0.1% SDS after 5 min at 95 °C followed by gradual cooling. The experimental record reports single-stranded concentrations of 1,688.51–1,709.40 ng/µL. After hybridization, ScanDrop 260/280 ratios were 1.92 for A and 1.88 for B; Bioanalyzer sizes were 129 bp and 173 bp. These apparent sizes exceed the 123-bp and 156-bp duplex references and should be interpreted within the sizing limitations of the assay rather than as base-resolved sequence evidence.

pET-21a(+) was double-digested with NdeI and XhoI, gel-purified and ligated to each hybridized insert at a nominal 3:1 insert:vector molar ratio. Ligation products were transformed into E. coli DH5α and selected on ampicillin. Colony PCR and capillary sequencing were performed as described in the source experimental record. The present software paper reuses these physical observations as a case study but does not claim expression, purification, oxidative refolding, receptor activity or therapeutic equivalence.

4.4 Post-sequencing validation by G-Synth

Validation was restricted to the four author-designated correct files: A Forward Seq.ab1, A Reverse Seq.ab1, B Forward Seq.ab1 and B Reverse Seq.ab1. No other chromatogram was admitted. The reads were analysed against the 123-bp A and 156-bp B cassette references. Although the filenames used the .ab1 extension, binary magic-number inspection identified SCF content; G-Synth therefore parsed the files by content rather than extension. For each chain, the forward and reverse read were independently placed, reverse-complemented when required, clipped to the reference and merged position by position into a reference-guided consensus. Consensus coverage denotes positions with at least one oriented call; bidirectional overlap denotes positions called from both orientations; overlap agreement denotes the fraction of those positions with identical F/R calls.

Because the source experiment did not prespecify a trace-quality threshold, consensus coverage, identity and overlap agreement were the primary retrospective endpoints; a fixed-threshold quality analysis was retained only as supplementary sensitivity information.

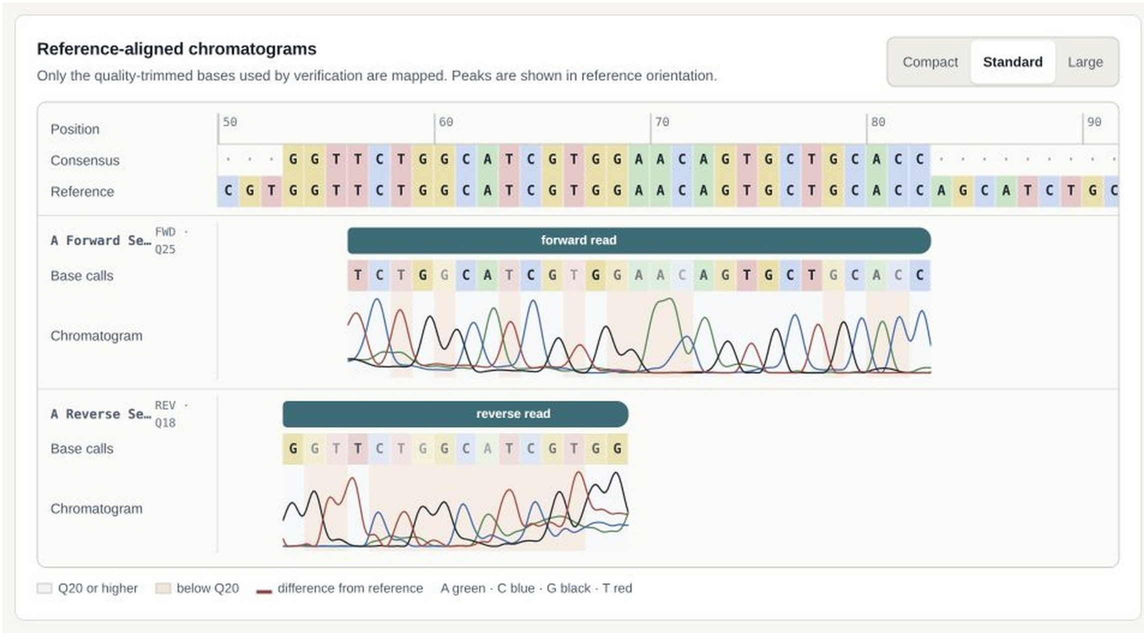
Chain	Reference	Consensus coverage	Consensus identity	Consensus differences	F/R overlap	F/R agreement	Independent coverage
A	123 bp	100.0%	100.0%	0	56.1%	100.0%	100.0%
B	156 bp	100.0%	100.0%	0	71.2%	100.0%	100.0%

The assembled consensus covered 100% of both references with 100% identity and no consensus differences. A contained 56.1% bidirectional overlap and B contained 71.2%; all overlapping F/R calls agreed (100%). Thus the paired reads jointly span each complete insert reference, while the overlap is independently observed on both orientations. Independent local alignments in Biopython 1.87 reproduced 100% combined coverage and 100% identity across aligned bases. The Geneious Prime figures from the experimental study reported visually concordant alignments but were not re-executed in the scripted environment.

Scientific interpretation

The oriented F/R assembly establishes complete consensus coverage, 100% identity and perfect agreement wherever the two reads overlap. This validates the intended A- and B-chain insert sequences at consensus-call level. It does not imply that every position was independently observed from both orientations, and the insert-only references do not test both vector–insert junctions; a prospective validation study should address those additional endpoints with predefined sequencing criteria.

A Reference-aligned A-chain chromatograms



B Reference-aligned B-chain chromatograms

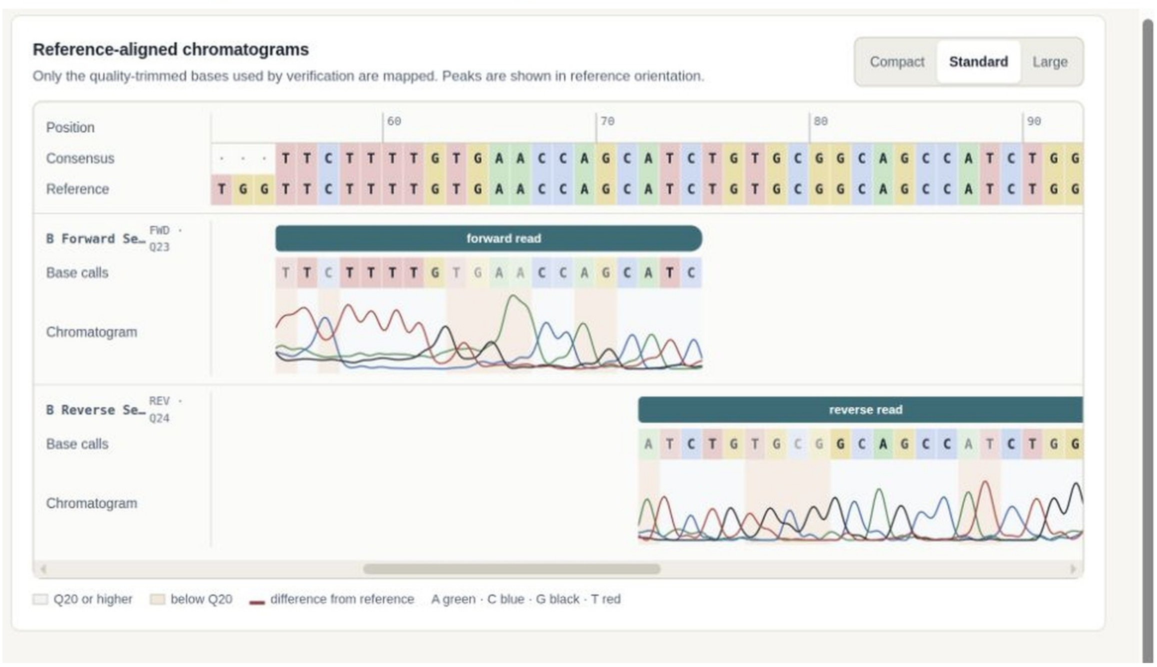


Figure 6. G-Synth's interactive reference-aligned chromatogram view for representative approved A- and B-chain reads. Base calls, consensus, quality context and four-channel peaks are displayed in reference orientation.

Alt text: G-Synth's interactive reference-aligned chromatogram view for representative approved A- and B-chain reads. Base calls, consensus, quality context and four-channel peaks are displayed in reference orientation.

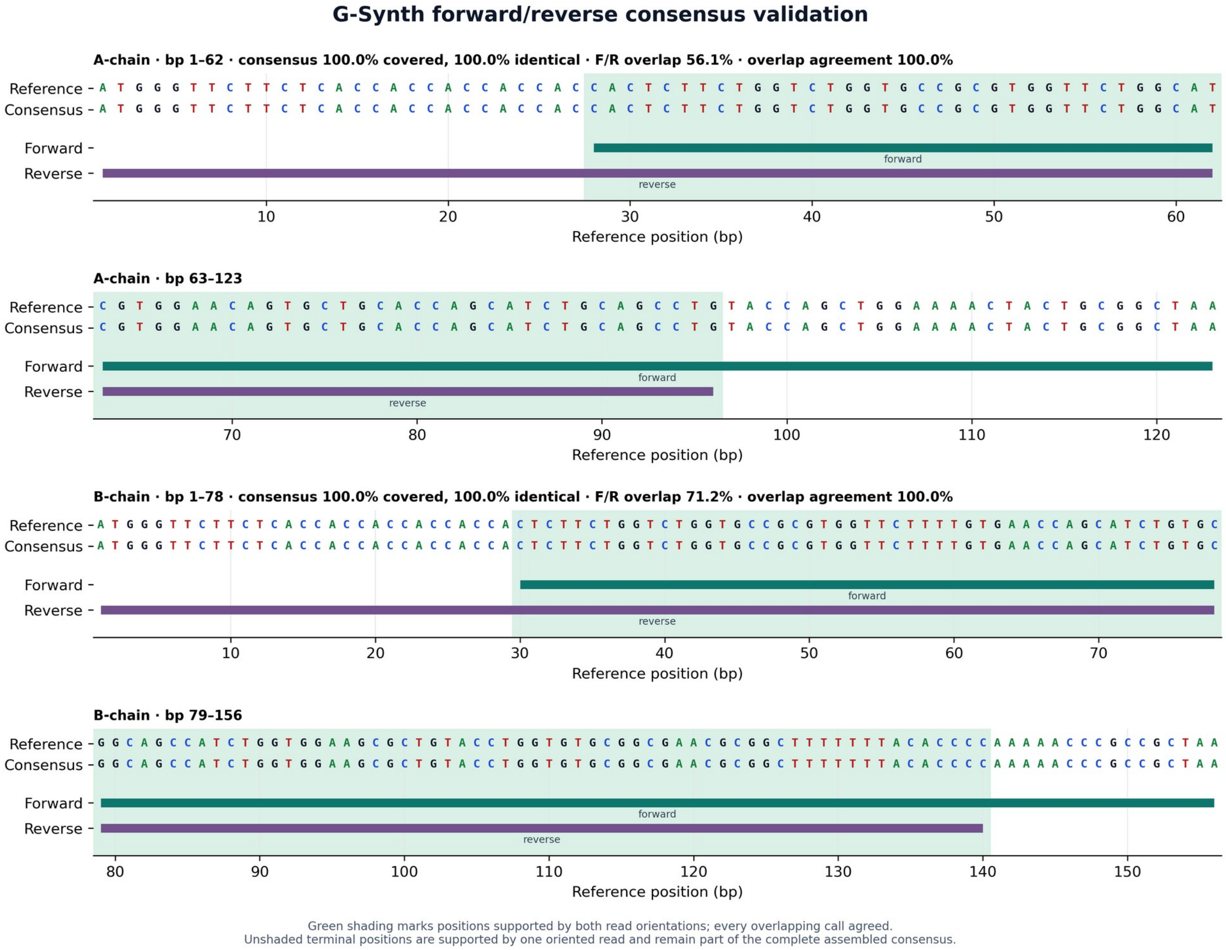


Figure 7. G-Synth F/R consensus validation from the four author-designated chromatograms. The oriented reads jointly cover 100% of both references with 100% consensus identity. Green shading marks bidirectional overlap (56.1% for A and 71.2% for B); every overlapping F/R call agreed.

Alt text: G-Synth F/R consensus validation from the four author-designated chromatograms. The oriented reads jointly cover 100% of both references with 100% consensus identity. Green shading marks bidirectional overlap (56.1% for A and 71.2% for B); every overlapping F/R call agreed.

5. Relationship to existing software

The comparison below separates G-Synth's precise contribution from adjacent functionality. GeneDesign and DNA Chisel focus on synthetic-sequence design and optimization [61–63]; j5, basicsynbio and DNA-BOT address assembly planning or build automation [58–60]; Geneious is a broad commercial environment [65]; DIVA, CloneCoordinate and InSillyClo cover increasingly integrated construction workflows [69–71]; and Biopython supplies reusable analysis primitives [66]. G-Synth is narrower than several of these systems. Its contribution is the coupled SSD/ESD synthesis-order model, exact two-strand release gate and continuity into post-Sanger consensus.

Tool/category	Primary scope	Relationship to SSD/ESD claim
G-Synth	SSD/ESD order molecules, restriction geometry, exact reconstruction, cloning and F/R consensus	Implements the claimed workflows and blocks export when molecular invariants fail
GeneDesign / DNA Chisel	Synthetic-gene design and sequence optimization	Foundational prior art for design automation; no identical SSD/ESD release gate identified

Tool/category	Primary scope	Relationship to SSD/ESD claim
j5 / basicsynbio / DNA-BOT	Assembly design, standardized parts and laboratory build instructions	Broader or method-specific assembly automation; not the same synthesis-order abstraction
Geneious Prime	Broad commercial cloning, annotation and sequence analysis	Secondary qualitative comparator; no automated head-to-head benchmark was executed
DIVA / CloneCoordinate / InSillyClo	Integrated construction planning, coordination or Golden Gate/MoClo campaigns	Directly limits any general end-to-end novelty claim; distinct from the exact SSD/ESD gate
Biopython	Programmatic parsing, alignment and molecular-biology primitives	Independent executable numerical comparator; equivalent logic can be scripted

This table is a workflow-positioning comparison, not a speed or accuracy benchmark. No proprietary Geneious algorithm was reverse-engineered, and no claim is made that G-Synth has broader functionality. Instead, the experimental Geneious views provide an independent qualitative check, while the numerical cross-check uses openly scriptable Biopython alignment. A future prospective benchmark should use blinded constructs, identical trimming thresholds and predefined acceptance criteria across platforms.

6. Discussion

The insulin case study separates three evidential levels. First, the current protein-input workflow demonstrates that G-Synth can preserve the mature A- and B-chain amino-acid sequences through E. coli-conditioned reverse translation, SSD, duplex reconstruction and restriction-cloning simulation. Second, exact regeneration of the four experimental order molecules establishes retrospective agreement between G-Synth's design logic and the molecular plan that reached the bench. Third, F/R assembly yields complete consensus coverage, 100% identity and 100% agreement across the bidirectional overlaps for those tested constructs. Positions outside the overlaps are supported by one oriented read rather than two, a distinction the software reports explicitly instead of obscuring within a single coverage value. The physical data validate the experimentally synthesized sequences, not the newly generated synonymous codon choices.

Restriction-aware design is particularly vulnerable to plausible-looking errors because recognition sites are visually familiar while cleavage products are strand-specific. Storing cut positions and calculating remainders avoids pair-specific tail constants. Reading the ends back from the reconstructed molecule adds an independent internal check. The same philosophy applies to primer extensions and virtual gels: G-Synth shows what the model establishes and labels what remains experimental.

Several limitations define the scope. First, long-oligonucleotide chemical synthesis can produce truncations and heterogeneous products; computational exactness does not predict synthesis yield or purification success. Second, orthogonal internal overhangs reduce one class of misassembly but do not model ligation kinetics. Third, restriction-cloning simulation does not predict methylation sensitivity, star activity, buffer compatibility, partial digestion or topology-dependent migration. Fourth, codon optimization and ORF integrity do not establish expression, folding, disulfide formation or insulin receptor activity. Separate production of A and B chains requires validated purification, cleavage, oxidative assembly, structural characterization and bioactivity studies [6–9,35,36,45–47,50,54–57]. Finally, the retrospective traces require prospective confirmation with predefined quality criteria and outside-vector primers spanning both junctions.

The appropriate next validation step is a preregistered, prospective design-to-sequence study. At least three independent constructs should include the NdeI/XhoI glargine path, a mixed 5′/3′ cohesive-end path and an internal-site negative control. Required records should include exported oligos, enzyme and vector lots, uncut/single-cut/double-cut controls, colony screening, raw capillary files, full junction and insert coverage, and independent review. User-centred validation with bench scientists should be reported separately from algorithmic correctness.

7. Conclusions

G-Synth formalizes Small Sequence Design and Extended Sequence Design as auditable synthesis-order workflows and carries a digital molecule from peptide input through host-selected reverse translation, oligonucleotide ordering, exact reconstruction, explicit hybridization, staged ligation, editable annotation, virtual PCR/digest/gel experiments and quality-aware post-sequencing review. The current insulin glargine demonstration began with the mature A- and B-chain proteins and passed every computational gate through pET-21a(+) cloning. In the separate retrospective track, SSD exactly regenerated all experimentally synthesized A/B oligonucleotides and assembled the approved F/R chromatograms into 100%-covered, 100%-identical consensus sequences with perfect agreement in their bidirectional overlaps. The software passed 1,470 automated tests across its engine, API and interface. The central contribution is not generic end-to-end software, but the first disclosed automation, to our knowledge, of the specifically defined SSD/ESD two-strand release gate and its continuity into post-sequencing evidence.

8. Materials and Methods

8.1 Reproducible software analyses

The current protein-to-clone workflow and peptide-start decisions were executed with tools/publication/validate_peptide_and_enzyme_logic.py. The script records both amino-acid inputs, the selected host profile, constraints, optimized coding sequences, translations, SSD molecules, duplex equality, cohesive ends, cloned-product lengths, junctions and hashes in publication_evidence/peptide_and_enzyme_validation.json. The retrospective design and cloning case study was executed with tools/publication/design_glargine_case_study.py. That script records the experimental-source-record SHA-256 hash, regenerates the oligonucleotides, asserts exact equality with the recorded synthesis molecules, translates the cassette, simulates pET-21a(+) cloning and writes publication_evidence/glargine_ab_design_and_cloning.json. Trace reanalysis used tools/publication/validate_insulin_correct_traces.py, which admits only the four author-designated files, hashes every reference and trace, constructs oriented G-Synth F/R consensus sequences, quantifies bidirectional overlap and agreement, performs an independent Biopython 1.87 local alignment and writes a JSON evidence record plus figure. Full commands are supplied in the Supporting Information.

The codon profiles were reconstructed with tools/update_hive_codon_tables.py from FDA HIVE service object 537 and committed as gsynth_engine/data/codon_usage_hive_2021.json. The updater requests genomic species data with descendant taxa, prefers RefSeq and falls back to GenBank only for K. phaffii and N. benthamiana, validates the returned taxon, requires exactly 64 non-negative codon counts and checks that their sum equals the reported total. The application divides each raw count by the maximum count among synonymous codons for that amino acid. The shipped September 2021 snapshot was retrieved on 2 September 2026 and is identified by SHA-256 in the API. Because these are species-wide genomic profiles rather than prespecified highly expressed genes, the geometric-mean score is reported as profile-relative CAI. No expression-yield inference is made. The machine-readable validation record publication_evidence/codon_host_profile_validation.json confirms 15 complete and numerically distinct 64-codon weight vectors and preservation of a 100-residue benchmark protein under every profile.

Peptide-start and restriction-catalogue checks were regenerated with tools/publication/validate_peptide_and_enzyme_logic.py. The resulting publication_evidence/peptide_and_enzyme_validation.json records four automatic and overridden peptide-role cases, host-wise reverse translation of the mature glargine A chain, translation hashes, the 109/289 geometry-to-name inventory and the independently derived HindIII 5′-AGCT end. The mature chain remained unchanged and did not receive an artificial ATG under all 15 host profiles; the complete-ORF override added exactly one initiator when required.

8.2 Computational hybridization and staged ligation

Both hybridization inputs were normalized as unambiguous DNA written 5′→3′. The second was reverse-complemented for offset evaluation, and every ungapped antiparallel placement was scored as twice the number of complementary columns minus three times the number of mismatches. Deterministic ties favoured more pairs, fewer mismatches, longer overlap, fewer terminally unpaired bases and the placement closest to flush. Internal gaps were not introduced because they represent bulges rather than terminal cohesive ends. Unpaired terminal intervals were classified by strand, side, polarity and sequence; overlaps shorter than four complementary bases were reported as insufficient. Perfect overlaps were evaluated with the implemented SantaLucia nearest-neighbour and Owczarzy salt-correction model under recorded buffer conditions; mismatched overlaps were not assigned a melting temperature.

For cloning handoff, the inherited duplex, enzyme pair and provenance were transferred without rewriting the strands. The vector was independently cut at the two selected unique sites. G-Synth compared the polarity and reverse-complement sequence of both insert–vector end pairs, then evaluated duplex pairing, cut uniqueness, orientation, regenerated sites and coding frame. Recombinant maps and downstream virtual analyses were enabled only after an explicit in-silico ligation state transition.

8.3 Biological inputs, cassette design and G-Synth analysis

Insulin glargine A- and B-chain amino-acid sequences were retrieved from DrugBank record DB00047 (database version 5.1.10). In the current workflow, the chain sequences were entered directly as peptides. Automatic mode classified both non-methionine N termini as mature peptide/fusion inserts. G-Synth used its E. coli HIVE-CUTs/CoCoPUTs profile, retained TAA termination and optimized under 40–60% local GC, five-base homopolymer, 15-nt repeat, low-frequency-codon and internal NdeI/XhoI constraints. Every resulting DNA was re-translated before SSD. Each cassette comprised the NdeI-provided initiator, MGSSHHHHHHSSGLVPRGS (6×His tag, linkers and thrombin-recognition segment), the G-Synth-optimized chain and TAA. SSD derived the NdeI- and XhoI-compatible strand products from cut geometry, generated both order molecules, re-annealed them in silico, verified exact reconstruction on both strands, translated the reconstructed coding sequence and simulated directional insertion into pET-21a(+).

The source experimental study used GenScript reverse translation and VectorBuilder codon optimization for E. coli. Those coding sequences were preserved unchanged in a second analysis track because they, rather than the new G-Synth synonymous designs, were physically synthesized. G-Synth repeated SSD, hybridization, translation and cloning on the experimental inputs and required exact equality with the recorded order molecules. Benchling and ExPASy operations are reported as experimental provenance; the present primary validation was performed with G-Synth.

8.4 Oligonucleotide synthesis, deprotection and purification

Forward and reverse A- and B-chain oligonucleotides were synthesized at the ESSBO Genomics Technology Platform by phosphoramidite chemistry on a MerMade 4 synthesizer (LGC Biosearch Technologies, Hoddesdon, UK) using Universal Support Columns, DMT-off (N-iPr), 1000 Å and 1 µmol scale. The source program comprised acetonitrile initialization washes; two deblocking deliveries of 3% trichloroacetic acid in dichloromethane; activator/amidite coupling in two subinjection stages; acetonitrile washing; Cap A and Cap B deliveries; iodine oxidation; final deblocking and washing; and cleavage/deprotection in 1 mL concentrated ammonia for 18 h at 70 °C. All volumes, durations and equalization times are reproduced without omission in Table S5.

Purification used an equal volume of butanol prechilled to −20 °C, 30 s vortexing, 30 min at −20 °C and centrifugation at 14,000 rpm for 15 min at 4 °C. Pellets were washed twice with 1 mL of 70% ethanol at −20 °C; each wash was followed by 10 min at 14,000 rpm and 4 °C. Pellets were air-dried for 10–15 min and resuspended in ultrapure sterile water. Purity and concentration were assessed by ScanDrop spectrophotometry (Analytik Jena) and 2% agarose gel electrophoresis at 100 V for 45 min. Rotor radius was not recorded; consequently rpm cannot be converted reproducibly to relative centrifugal force.

8.5 Hybridization and analytical quality control

The 20× SSC stock contained 3 M NaCl and 0.3 M sodium citrate and was adjusted to pH 7.0. For each duplex, 10 µL of each purified complementary strand was combined with 20 µL of 2× SSC containing 0.1% SDS, heated at 95 °C for 5 min and cooled gradually to room temperature over 30 min. Samples were stored at 4 °C for short-term use or −80 °C for long-term storage. Duplex concentration, purity and apparent size were assessed by ScanDrop, 2% agarose electrophoresis (100 V, 45 min) and an Agilent 2100 Bioanalyzer with DNA 1000 reagents (25–1000-bp stated range; ±5% sizing resolution; 15% quantitation CV for 100–500 bp).

8.6 Vector preparation, ligation, transformation and clone screening

pET-21a(+) was digested in 50 µL containing 1 µg vector, 1 µL NdeI (10 U/µL), 1 µL XhoI (10 U/µL), 5 µL 10× CutSmart buffer and nuclease-free water for 2 h at 37 °C. The digest was resolved on 1% agarose and gel-purified with an Invitrogen kit. Ligation used a 3:1 insert:vector molar ratio calculated with NEBioCalculator, 50 ng digested vector, the calculated insert quantity, 1 µL T4 DNA ligase (400 U/µL), 2 µL 10× ligase buffer and water to 20 µL; reactions were held at 16 °C for 2 h and 4 °C overnight. Ligation products were assessed by ScanDrop, Bioanalyzer and 2% agarose electrophoresis (100 V, 45 min).

Competent E. coli DH5α cells were heat-shocked for 50 s at 42 °C, recovered in SOC medium and plated on LB agar containing 100 µg/mL ampicillin. Colonies were screened by PCR using both T7 and insert-specific primer pairs. Positive clones were subjected to plasmid miniprep (Thermo Fisher Scientific) and capillary sequencing. LB medium and ampicillin were from Sigma-Aldrich; SOC medium and DH5α cells were from Thermo Fisher Scientific.

8.7 Capillary sequencing, G-Synth consensus and secondary confirmation

Each 20-µL BigDye Terminator v3.1 reaction contained 20 ng purified DNA, 2 µL BigDye Terminator v3.1, 3 µL 5× sequencing buffer and 1 µL forward or reverse primer at 3 µM. Cycling comprised 96 °C for 60 s followed by 35 cycles of 96 °C for 10 s, 50 °C for 5 s and 60 °C for 4 min. Products were ethanol-precipitated and analysed on an Applied Biosystems 3500 Genetic Analyzer with POP-7 polymer at the ESSBO Genomics Technology Platform.

Only A Forward Seq.ab1, A Reverse Seq.ab1, B Forward Seq.ab1 and B Reverse Seq.ab1 were admitted to the present analysis. Content signatures identified the files as SCF despite their extensions. G-Synth parsed calls, qualities, peak locations and four-channel traces; evaluated both orientations; transformed placed reads into reference orientation; and merged them position by position. Consensus calls were selected by supporting-read count and then cumulative quality; exact ties were emitted as N. Consensus coverage, identity, bidirectional overlap and overlap agreement were computed separately. Because the source experiment had no prespecified trace-quality acceptance threshold, no post hoc cutoff was used for the primary retrospective endpoint. Biopython PairwiseAligner supplied an independent executable numerical comparison, and the Geneious Prime 2024.0.7 figure from the same experiment supplied a secondary qualitative confirmation.

Data and Software Availability

The G-Synth source code, tests and documentation are available under the MIT License at https://github.com/Midotech31/g-synth-app. A versioned archival DOI will be inserted after repository release and Zenodo deposition. Reproducibility scripts, the experimental source record and JSON evidence are under tools/publication/ and publication_evidence/. Glargine records PQ362993 and PQ362994 are in GenBank. Raw capillary files contain no human data; deposition details will be finalized before submission or supplied confidentially to reviewers when permitted.

Supporting Information

Complete designs and oligonucleotides; algorithmic invariants and executable commands; experimental and QC records; trace manifest; interface evidence; comparison data; and release checklist (DOCX and PDF).

Author Contributions

M.M. conceived G-Synth and the PCR-free construct strategy, defined the scientific requirements, supervised software development and validation, supplied and adjudicated the experimental evidence, and drafted the manuscript. Z.Y.Z. contributed experimental investigation and manuscript review. Y.S., A.I.H., M.A., K.B., H.S.B. and S.B. contributed experimental investigation. D.S. contributed supervision, experimental review and manuscript review.

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Generative-AI use

OpenAI Codex (GPT-5 family, Codex desktop application; accessed June–September 2026) assisted under Mohamed Merzoug's direct supervision with software review and refactoring, test scaffolding and execution, documentation and manuscript drafting, literature-discovery assistance, copy-editing, release preparation and document-layout quality control. The authors supplied and interpreted the experimental evidence, adjudicated the biological assumptions, reviewed and edited the AI-assisted output and retain responsibility for the accuracy, originality, licensing and scientific claims. Generative AI was not treated as an author, did not generate experimental observations and did not independently determine novelty or scientific validity.

Conflict of Interest

The authors declare no competing financial interest.

Figure Legends and Alt Text

Figure 1. G-Synth workflow and software architecture. The design, build, clone and verify stages share a deterministic Python engine. The API stores project provenance and the web workspace renders the engine outputs without reimplementing biological calculations.

Alt text: G-Synth workflow and software architecture. The design, build, clone and verify stages share a deterministic Python engine. The API stores project provenance and the web workspace renders the engine outputs without reimplementing biological calculations.

Figure 2. G-Synth hybridization view after direct transfer from SSD. Both order molecules remain entered 5′→3′; the lower strand is drawn antiparallel. The 123-bp core is completely paired, while the NdeI-derived 5′-TA and XhoI-derived 5′-TCGA products remain visibly unpaired as cohesive ends.

Alt text: G-Synth hybridization view after direct transfer from SSD. Both order molecules remain entered 5′→3′; the lower strand is drawn antiparallel. The 123-bp core is completely paired, while the NdeI-derived 5′-TA and XhoI-derived 5′-TCGA products remain visibly unpaired as cohesive ends.

Figure 3A. Pre-ligation compatibility gate for the current glargine A design. G-Synth shows the 125-bp insert, the NdeI and XhoI vector–insert junctions, their opposite polarities and all six passed checks while labelling the joined sequence as expected and withholding recombinant-product analyses.

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Figure 3B. Post-ligation state for the current glargine A-chain construct. A green confirmation panel reports the 5,490-bp recombinant product and shows both closed nucleotide junctions only after explicit in-silico ligation; the plasmid map, diagnostic digest, gel and exports then become available.

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Figure 4. G-Synth design view for the glargine A-chain input. The release gate reports exact two-strand reconstruction, terminal-end compatibility and the NdeI start-codon note before displaying the orderable construct.

Alt text: G-Synth design view for the glargine A-chain input. The release gate reports exact two-strand reconstruction, terminal-end compatibility and the NdeI start-codon note before displaying the orderable construct.

Figure 5A. G-Synth coordinate-level view of the insulin glargine A-chain recombinant pET-21a(+) construct. Editable feature tracks, codon-aligned translation and the regenerated NdeI and XhoI junctions are shown across the circular origin.

Alt text: G-Synth coordinate-level view of the insulin glargine A-chain recombinant pET-21a(+) construct. Editable feature tracks, codon-aligned translation and the regenerated NdeI and XhoI junctions are shown across the circular origin.

Figure 5B. G-Synth coordinate-level view of the insulin glargine B-chain recombinant pET-21a(+) construct, using the same editable annotation and strand-aware coordinate system.

Alt text: G-Synth coordinate-level view of the insulin glargine B-chain recombinant pET-21a(+) construct, using the same editable annotation and strand-aware coordinate system.

Figure 6. G-Synth's interactive reference-aligned chromatogram view for representative approved A- and B-chain reads. Base calls, consensus, quality context and four-channel peaks are displayed in reference orientation.

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